

Foundation Mathematics 1017SCG
Week 3 Workshop Questions

1. Convert the following to radians, leaving your answer as an exact value.
 - (a) 30°
 - (b) 45°
 - (c) 60°
 - (d) 90°
 - (e) 210°
 - (f) 300°
2. Convert the following to degrees. Where necessary, round your answer to two decimal places.
 - (a) $\frac{2\pi}{3}$ radians
 - (b) $\frac{5\pi}{6}$ radians
 - (c) $\frac{5\pi}{4}$ radians
 - (d) $\frac{3\pi}{2}$ radians
 - (e) π radians
 - (f) 2 radians
3. Complete the following table giving exact values.
(Use a combination of the special triangles and the unit circle).

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
30°			
45°			
120°			
240°			
315°			

4. Complete the following table giving exact values.
(Use a combination of the special triangles and the unit circle).

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
$\frac{\pi}{3}$			
$\frac{3\pi}{4}$			
$\frac{4\pi}{3}$			
$\frac{11\pi}{6}$			

5. Complete the following table giving exact values.

θ	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
90°			
π			
2π			
$-\frac{\pi}{4}$			
$-\frac{5\pi}{6}$			

6. If possible, simplify the following.

(a) $3x + 5x$

(b) $9x^2 - 2x^2$

(c) $2x + 4x + 5y$

(d) $6x^2 - 5x^2 + 9x$

(e) $5x^2 - 4x + 8$

(f) $9x - 5y + 3x + 4y$

7. Consider the expression $10a^3 - 6a^2 + 4a + 13$

(a) How many terms in the expression?

(b) What is the constant term?

(c) What is the coefficient of a^2 ?

(d) What is the value of the expression given $a = 2$?

(e) What is the value of the expression given $a = -1$?